

The Unique Games Theorem

An expository account of the proof

A rewrite, for human readers, of the manuscript *The Unique Games Theorem*
(OpenAI, September 23, 2026)

Prepared with Claude (Anthropic) · October 7, 2026

Abstract

In September 2026 OpenAI posted a manuscript, written by an internal model, that proves Khot’s Unique Games Conjecture by a deterministic polynomial-time reduction from 3SAT. The manuscript is complete but compressed: definitions, estimates and parameter choices arrive in the order the proof needs them, with little explanation of why they are there. This note presents the same proof for a reader who knows the PCP theorem, long codes and the outline of the 2-to-2 Games theorem but has not read the manuscript. It explains what each step is for, does every key calculation in full, gives complete proofs of the one genuinely new ingredient (a “latent alphabet gadget”), and sketches the parts that adapt known machinery, with pointers to the original. It is an exposition, not a referee report; [Section 12](#) lists what was and was not checked and where a careful reader should concentrate.

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1 Introduction

1.1 The statement

A *Unique Games instance* consists of a finite set of vertices, a finite alphabet K , and a list of constraints $e = (u_e, v_e, \pi_e)$, where each π_e is a permutation of K . A labeling λ satisfies e when $\lambda(v_e) = \pi_e(\lambda(u_e))$; the value $\text{val}(G)$ is the largest fraction of constraints a labeling can satisfy. Every label at one end of a constraint allows exactly one label at the other end, which is what makes a perfectly satisfiable instance easy: fix one label per connected component and propagate. The Unique Games Conjecture [8] says that *nearly* satisfiable instances are nevertheless hard.

Theorem 1.1 (Unique Games Theorem; Theorem 1.1 of [1]). *For all fixed $\varepsilon, \delta \in (0, 1/2)$ there are an integer $s \geq 1$ and a deterministic polynomial-time reduction mapping a 3CNF formula φ to an explicit unweighted Unique Games instance G_φ over the alphabet $K = \mathbb{F}_2^s$ such that*

$$\varphi \text{ satisfiable} \Rightarrow \text{val}(G_\varphi) \geq 1 - \varepsilon, \quad \varphi \text{ unsatisfiable} \Rightarrow \text{val}(G_\varphi) \leq \delta.$$

The constraint graph is simple and bipartite, and every constraint is a translation $\lambda(v_e) = \lambda(u_e) + c_e$ in $\mathbb{F}_2^s$.

This is the usual form of the conjecture: both errors are fixed first, and the alphabet is allowed to depend on them. Through the reductions of Khot–Kindler–Mossel–O’Donnell [9], Khot–Regev [12], Raghavendra [16] and others, the theorem makes the Goemans–Williamson ratio for Max-Cut, factor 2 for Vertex Cover, and the basic-SDP threshold for every Max-CSP NP-hard to beat. [Section 11](#) lists these.

1.2 Where things stood, and what was missing

The 2-to-2 Games theorem of Khot, Minzer and Safra [10, 11], built on Dinur–Khot–Kindler–Minzer–Safra [4] and Barak–Kothari–Steurer [3], gives NP-hardness of distinguishing Unique Games of value about $1/2$ from value δ . Two facts frame the remaining problem.

1. **The soundness side is not the problem.** Because the parallel-repetition bounds of Rao [15] and Dinur–Steurer [5] for projection games do not depend on the alphabet size, it suffices to prove hardness of distinguishing value $\geq 1 - \varepsilon$ from value $\leq 1 - c$ for one *fixed* $c > 0$ and *every* $\varepsilon > 0$. Repeating $t = t(c, \delta)$ times drives $1 - c$ below δ while completeness drops only to $1 - t\varepsilon$, and ε can be chosen after t . ([Section 2.1](#) does this calculation.)
2. **The completeness side is the problem.** The shortcode test that underlies the 2-to-2 theorem has an honest pass probability of only about $1/2$ once it is forced to be a permutation test. [Section 3.1](#) explains why.

So what is needed is a test whose honest strategies pass with probability $1 - p_*$ for arbitrarily small p_* , while every labeling that passes with probability 0.99 can still be decoded. The manuscript does exactly this.

1.3 The proof on one page

1. **Start** from Håstad’s 3LIN gap [7]: weighted systems of parity equations on three variables, value $\geq 1 - \xi$ versus $\leq 1/2 + \xi$. View it as the usual equation-versus-variable projection game, which has value $\leq 14/15$ in the NO case (Section 2.2).
2. **Inner code.** An answer to a tuple U of k equations is a vector $z \in \mathbb{F}_2^m$ ($m = 1 + 2k$, homogeneous coordinates). In the Barak–Kothari–Steurer (BKS) shortcode, z is encoded by the function $M \mapsto Mz$ on $\ell \times m$ binary matrices, and the test compares M with a rank-one perturbation $M + al^\top$. Honest functions change with probability about $1/2$.
3. **The new idea: a nonlinear decoder.** Replace the target space $K = \mathbb{F}_2^\ell$ by a much larger space $\mathcal{V} \supseteq K$ and encode z by $P \mapsto C(Pz)$, where $P: \mathbb{F}_2^m \rightarrow \mathcal{V}$ is linear and $C: \mathcal{V} \rightarrow K$ is a fixed *nonlinear* map that commutes with translations by K . Perturb P by $al^\top$ with a drawn from a special noise law μ on $\mathcal{V}$. The gadget lemma (Section 3) builds C and μ so that
 - C almost never notices the noise: $\Pr[C(x + a) \neq C(x)] \leq p_*$;
 - every family of linear functionals whose restriction to K has rank at least a threshold r_* notices the noise with probability $\geq 1/8$;
 - and, crucially, r_* depends only on p_* , not on $\ell = \dim K$.

Honest tables therefore pass with probability $\geq 1 - p_*/2$.

4. **Soundness, step 1: back to the ordinary test.** In the Fourier expansion over tables, the new noise multiplies each coefficient by a number in $[0, 1]$ that is at most $7/8$ on “high-rank” frequencies. So a labeling passing with probability 0.99 has most of its Fourier mass at rank $< r_*$, and therefore passes the *ordinary* BKS equality test with probability at least about 2^{-r_*} , a constant independent of ℓ (Section 6).
5. **Soundness, step 2: structure.** The inverse shortcode theorem, derived from the Khot–Minzer–Safra Grassmann expansion theorem [11], says such a function agrees with an affine function $Mz + u$ on a slice cut out by a bounded number r_s of row and column conditions (Section 7).
6. **Soundness, step 3: decoding.** As in the 2-to-2 proof, this structure is turned into strategies for a smooth equation-versus-variable game in which both players receive a little shared linear “advice”. The strategies agree with probability $\gamma > 0$, independent of the tuple length k (Section 8).
7. **Soundness, step 4: contradiction.** On many coordinates (“clean” ones) the advice says nothing, and there the players are playing the original game in parallel. Parallel repetition bounds their agreement by $\exp(-ck^{1/3})$, which is below γ for large k (Section 9).
8. **Finish.** The resulting game has completeness $1 - k\xi - p_*/2$ and soundness 0.99 . Remove weights, make it simple and bipartite, and apply t -fold parallel repetition (Section 10).

1.4 What is new and what is inherited

Most of the soundness argument adapts known machinery: Håstad’s 3LIN hardness, the BKS matrix formulation of the shortcode, the KMS Grassmann expansion theorem, the smooth outer game with shared linear advice of Khot–Minzer–Safra and Minzer–Zheng [10, 14],

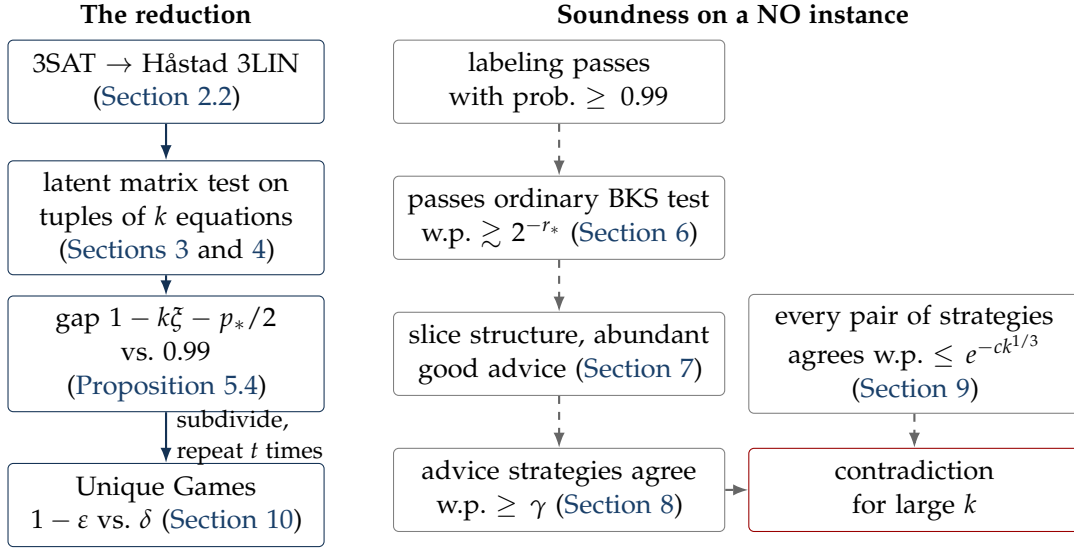


Figure 1: Roadmap. Solid arrows: the reduction. Dashed arrows: the soundness argument, which plays a decoding lower bound against a repetition upper bound.

Khot–Safra’s “virtual tables” and Fourier list decoding [13], folding in the style of Bellare–Goldreich–Sudan [2], and Dinur–Steurer repetition [5]. The new ingredients are:

- **The latent alphabet gadget** (Section 3), a nonlinear, translation-equivariant map that is stable under a noise that every high-rank linear observer detects. This is what breaks the completeness barrier. Its proof is elementary (finite fields, a polynomial-identity argument, and a potential function) and is given here in full.
- **Exact table keys** (Section 4.2): tables on a tuple and on its sparse projection are identified exactly, so the projected player’s answers come from the same labeling and no separate consistency test is needed.
- **An erasure argument** (Section 7.2) showing that the structure promised by the inverse theorem is visible from *random* advice with constant probability, not merely that it exists somewhere.
- The realization, visible in the model’s released reasoning summary, that a *fixed* soundness gap with completeness tending to 1 suffices, which makes a constant-factor damping of high-rank Fourier mass enough.

1.5 Formal verification

The repository includes a Lean 4 formalization of this result (directory `lean/OAI/Computability/UniqueGames`, about 480 files and 187,000 lines) and a Comparator challenge file `UniqueGamesTheorem.lean` that states the theorem using only Mathlib. That statement asserts, for real $0 < \varepsilon, \delta < 1/2$, the existence of a reduction from the binary-encoded language of satisfiable 3CNF formulas, computable in polynomial time by a Mathlib TM2 Turing machine, to simple bipartite translation instances over $\mathbb{F}_2^s$ with completeness $1 - \varepsilon$ and soundness δ . The permitted axioms are the three standard ones (`propext`, `Quot.sound`,

Classical.choice). The file names indicate that the external inputs (Håstad’s verifier, the KMS expansion theorem, projection-game repetition) are formalized rather than assumed. I read the challenge statement but did not run the check, which requires building the whole development.

1.6 How this note is organized

Section 2 reduces the theorem to a fixed-gap statement. Section 3 builds the gadget. Section 4 defines the Unique Games instance and proves completeness. Section 5 states the two bounds that make up soundness, and Sections 6 to 9 prove them. Section 10 assembles the parameters. Section 12 is a guide for checking the proof, with a table matching this note to the manuscript. Throughout, vector spaces are over $\mathbb{F}_2$ unless stated otherwise, E^* denotes the dual of E , and “uniform” sampling from a vector space includes 0.

2 Two easy reductions

2.1 A fixed gap is enough

The whole proof aims at a weighted translation game with completeness arbitrarily close to 1 and soundness at most 0.99. This subsection shows why that suffices. Two standard tools are used.

Theorem 2.1 (Projection-game repetition; Dinur–Steurer [5, Cor. 1.2]). *Let G be a finite bipartite projection game with any probability distribution on its constraints. If $\text{val}(G) \leq 1 - g$, then for every $n \geq 1$, $\text{val}(G^{\otimes n}) \leq (1 - g^2/16)^n$. The bound does not depend on the alphabet or on the number of questions.*

A unique game is a projection game, and the n -fold product of translation constraints with offsets $c_1, \dots, c_n$ is the translation by $(c_1, \dots, c_n)$ on K^n , so repetition stays inside the class of translation games.

Lemma 2.2 (Rounding and subdivision; Lemma 7.1 of [1]). *Let H be a permutation game with h constraint occurrences carrying positive rational weights summing to 1 (loops and parallel occurrences allowed). For rational $\tau > 0$ one can build in polynomial time an unweighted simple bipartite game B on the same alphabet, with $4\lceil h/\tau \rceil$ edges, such that*

$$\left| \text{val}(B) - \left(1 - \frac{1 - \text{val}(H)}{4} \right) \right| \leq \frac{\tau}{4}.$$

Translation constraints stay translation constraints.

Proof idea. Round the weights to multiples of $1/Q$ with $Q = \lceil h/\tau \rceil$ by largest remainders and replace each occurrence by that many copies; this changes the value by at most $h/(2Q) \leq \tau$. Then replace each occurrence $u \rightarrow v$ with constraint ρ by a path $u - p - q - r - v$ through three fresh vertices, with identity constraints on the first three edges and ρ on the last. A satisfied occurrence extends to four satisfied edges and a violated one to exactly three, giving $\text{val}(B) = 1 - (1 - \text{val}(H'))/4$ exactly. Putting the original vertices and the q ’s on one side and the p ’s and r ’s on the other makes B bipartite, and fresh interiors make it simple. $\square$

Corollary 2.3. *Suppose that for every rational $d > 0$ there is a polynomial-time reduction from 3SAT to weighted translation games H over some alphabet $\mathbb{F}_2^\ell$ (allowed to depend on d) with*

$$\text{YES: } \text{val}(H) \geq 1 - d, \quad \text{NO: } \text{val}(H) \leq 99/100.$$

Then Theorem 1.1 holds.

Proof. Fix ε, δ . Apply Lemma 2.2 with $\tau \leq 1/200$; in the NO case $\text{val}(B) \leq 1 - (1/100 - \tau)/4 \leq 1 - 1/800$. With $g = 1/800$, Theorem 2.1 gives $\text{val}(B^{\otimes t}) \leq (1 - 1/10,240,000)^t$, so choose t (depending only on δ) to make this at most δ . Only now choose $d \leq \varepsilon/t$ and $\tau \leq \min\{1/200, \varepsilon/t\}$. In the YES case $\text{val}(B) \geq 1 - (d + \tau)/4$, and labeling $B^{\otimes t}$ coordinatewise gives, by a union bound, $\text{val}(B^{\otimes t}) \geq 1 - t(d + \tau)/4 \geq 1 - \varepsilon/2$. The product of simple bipartite graphs is simple bipartite, and the alphabet is $\mathbb{F}_2^{\ell t}$. $\square$

The order of quantifiers is the whole point: t is fixed by δ alone, because the NO gap $1/800$ does not depend on anything, and only then is the completeness error chosen. Alphabet-independent repetition is essential here, since the alphabet $\mathbb{F}_2^\ell$ is chosen after t .

2.2 The outer game

Theorem 2.4 (Håstad [7, Thm. 5.4]). *For every rational $\xi > 0$ there is a polynomial-time reduction from SAT to weighted systems $\mathcal{E}$ of equations $x_{v_1} + x_{v_2} + x_{v_3} = b$ over $\mathbb{F}_2$ such that $\text{OPT}(\mathcal{E}) \geq 1 - \xi$ in the YES case and $\text{OPT}(\mathcal{E}) \leq \frac{1}{2} + \xi$ in the NO case.*

Later arguments need the three variables of each equation to be distinct.

Lemma 2.5 (Distinct positions; Lemma 4.2 of [1]). *One can transform $\mathcal{E}$ in polynomial time into $\mathcal{E}'$ with three distinct variables per equation, such that $\text{OPT}(\mathcal{E}') \geq \text{OPT}(\mathcal{E})$ and $\text{OPT}(\mathcal{E}') \leq \frac{1 + \text{OPT}(\mathcal{E})}{2} + \frac{3}{100}$. In particular, with $\xi < 1/100$ the NO case gives $\text{OPT}(\mathcal{E}') \leq 4/5$.*

Proof idea. Replace each variable by 100 clones and each equation by all ways of choosing clones for its positions (distinct clones where the same variable repeats), with equal conditional weights. Completeness is clear. For soundness, let $t_v \in [-1, 1]$ be the average of $(-1)^x$ over the clones of v . With clones chosen independently, an equation is satisfied with probability $\frac{1}{2}(1 + (-1)^b t_{v_1} t_{v_2} t_{v_3})$. Rounding each t_v to its sign gives an assignment, and every equation it violates has probability at most $1/2$, so the average is at most $(1 + \text{OPT})/2$. Conditioning on distinctness costs at most $3/100$. $\square$

From now on every equation has three distinct variables. The *equation-versus-variable game* $G_{\mathcal{E}}$ picks an equation e by weight and a position $j \in \{1, 2, 3\}$ uniformly. One player gets e and answers with one of the four satisfying assignments of e ; the other gets the j -th variable of e and answers with a bit; they win if the bits agree. The variable player's answers form a global assignment, which violates a $1 - \text{OPT}$ fraction of equations, and on each violated equation the equation player disagrees in at least one of three positions. Hence

$$\text{OPT}(\mathcal{E}) \leq \frac{4}{5} \implies \text{val}(G_{\mathcal{E}}) \leq 1 - \frac{1 - \text{OPT}}{3} \leq \frac{14}{15}. \quad (1)$$

3 Breaking the completeness barrier: the latent gadget

3.1 Why the shortcode test has completeness one half

In the BKS formulation of the degree-two shortcode, an answer $z \in \mathbb{F}_2^m$ is encoded by the function $f_z(M) = Mz$ on $\text{Mat}_{\ell,m} = \text{Hom}(\mathbb{F}_2^m, \mathbb{F}_2^\ell)$. The test picks a uniform matrix M and uniform $a \in \mathbb{F}_2^\ell$, $l \in \mathbb{F}_2^m$, and compares M with $M + al^\top$. Since

$$f_z(M + al^\top) = f_z(M) + a(l^\top z),$$

the two values differ exactly when $l^\top z = 1$ and $a \neq 0$, that is, with probability about $1/2$. BKS accept if $f(M + al^\top) \in \{f(M), f(M) + a\}$, which honest functions always pass; that is a 2-to-2 test. To get a *unique* test one must demand equality, and then honest functions pass only half the time. This is the source of the “ $1/2$ ” in the Unique Games corollary of the 2-to-2 theorem.

The trouble is that the noise $a(l^\top z)$ reaches the codeword at full strength. The manuscript’s idea is to make the codeword blind to most of the noise while the soundness analysis still sees all of it.

3.2 A nonlinear decoder

Replace $\mathbb{F}_2^\ell$ by a larger space $\mathcal{V}$ containing $K = \mathbb{F}_2^\ell$, let tables be linear maps $P: \mathbb{F}_2^m \rightarrow \mathcal{V}$, and encode z by

$$F(P) = C(Pz),$$

where $C: \mathcal{V} \rightarrow K$ is a fixed map. For folding we need $C(x + h) = C(x) + h$ for $h \in K$. Perturb P by $al^\top$ with $a \sim \mu$ for a noise law μ on $\mathcal{V}$. Then Pz is a uniform element of $\mathcal{V}$ independent of a and of the fair bit $l^\top z$, so the honest encoding changes with probability $\frac{1}{2} \Pr[C(x + a) \neq C(x)]$.

What does soundness need from μ ? In the Fourier expansion over tables (Section 6), a frequency is a tuple $g = (g_0, \dots, g_{m-1})$ of functionals $g_i \in \mathcal{V}^*$, and averaging over l shows that the noise multiplies its coefficient by

$$\Lambda_g = \Pr_{a \sim \mu} [g_i(a) = 0 \text{ for all } i].$$

For the reduction to the ordinary BKS test we want Λ_g bounded away from 1 whenever the functionals g_i , restricted to K , span a space of large dimension. That is the gadget’s second property.

Why C must be nonlinear. If C were linear with $C|_K = \text{id}$, take G to be the span of the ℓ coordinate functionals of C . Its restriction to K has full rank ℓ , and some $g \in G$ has $g(a) \neq 0$ exactly when $C(a) \neq 0$, that is, exactly when $C(x + a) \neq C(x)$. Stability and detection would be the same event. So any useful C is nonlinear, and the content of the lemma is that a nonlinear map can be much more stable than every linear observer of comparable rank.

Theorem 3.1 (Latent alphabet gadget; Lemma 3.1 of [1]). *For every $p_* > 0$ there is an integer r_* such that for every sufficiently large ℓ there are a binary space $\mathcal{V} \supseteq K = \mathbb{F}_2^\ell$, a map $C: \mathcal{V} \rightarrow K$ and a probability law μ on $\mathcal{V}$ with:*

- (i) equivariance: $C(x + h) = C(x) + h$ for all $x \in \mathcal{V}$, $h \in K$;
- (ii) stability: $\Pr[C(x + a) \neq C(x)] \leq p_*$ for independent uniform $x \in \mathcal{V}$ and $a \sim \mu$;
- (iii) detection: every subspace $G \leq \mathcal{V}^*$ with $\dim(G|_K) \geq r_*$ satisfies $\Pr_{a \sim \mu}[\exists g \in G : g(a) \neq 0] \geq 1/8$.

The threshold r_* is fixed before ℓ , and everything is computable when p_* is rational.

The construction has three layers: a quadratic block over a finite field (Section 3.3); a deep tree of blocks with a fixed output space $B = \mathbb{F}_{2^d}^3$ (Sections 3.5 and 3.6); and an enlargement from B to an arbitrary K (Section 3.7).

The idea in one paragraph: one life versus many lives. Think of a noise symbol injected at a random leaf of a deep tree and propagating upward. At each level the nonlinear output “absorbs” the perturbation with a small probability θ , so after s levels it survives with probability about $e^{-\theta s}$: one life, lost at rate θ . A rank- r family of linear observers behaves differently. At each level it can lose at most *one* dimension, and only with probability at most $3r\theta$. Measuring rank by the harmonic number $H_r = 1 + \frac{1}{2} + \dots + \frac{1}{r}$, the expected loss per level is at most about 3θ , so killing all r dimensions takes about $H_r / (3\theta) \approx (\ln r) / (3\theta)$ levels. Taking $s \approx H_{r_0} / (8\theta)$ levels, the nonlinear output survives with probability about $r_0^{-1/8}$, which is small when r_0 is large, while a family starting at rank r_0 survives with probability at least $1/2$. The factor $\ln r_0$ is the whole separation.

3.3 The quadratic block

Fix $d \geq 1$, put $q = 2^d$ and $F = \mathbb{F}_q$, and let $X = B = F^3$, regarded as binary spaces of dimension $b = 3d$. Identify X^* and B^* with F^3 through the trace pairing $z \mapsto (y \mapsto \text{Tr}(z \cdot y))$, where $z \cdot y$ is the F -bilinear dot product and $\text{Tr} = \text{Tr}_{F/\mathbb{F}_2}$. Recall that the trace pairing is nondegenerate, that $\text{Tr}(t^2) = \text{Tr}(t)$, and that every element of F has a unique square root. Define

$$Q(x_1, x_2, x_3) = (x_2x_3, x_1x_3, x_1x_2), \quad D(x, a) = Q(x + a) + Q(x) + Q(a),$$

and the block map

$$C_{\text{blk}}: X \oplus B \rightarrow B, \quad C_{\text{blk}}(x, y) = y + Q(x).$$

It is equivariant in the second (“vertical”) argument: $C_{\text{blk}}(x, y + h) = C_{\text{blk}}(x, y) + h$.

Update spaces. For each F -line $A = Fv \subseteq F^3$ let

$$D_A = \{y \in B : v \cdot y = 0\}, \quad U_A = \{(a, Q(a) + w) : a \in A, w \in D_A\} \subseteq X \oplus B.$$

There are $q^2 + q + 1$ lines; set $\theta = 1/(q^2 + q + 1)$.

Lemma 3.2. Each U_A is a binary subspace of dimension b . For $a \in A$ and any x , $D(x, a) \in D_A$, and $D(a, a') = 0$ for $a, a' \in A$. Under the trace pairing the annihilator of D_A is A . The spaces U_A together span $X \oplus B$.

Proof. The coordinates of $D(x, a)$ are $x_2a_3 + a_2x_3$, $x_1a_3 + a_1x_3$, $x_1a_2 + a_1x_2$. For $a = tv$ every term of $v \cdot D(x, a)$ appears twice, so it vanishes; and for $a, a' \in A$ each coordinate of $D(a, a')$ vanishes. So Q is additive on A , and $(a, w) \mapsto (a, Q(a) + w)$ is a linear bijection $A \oplus D_A \rightarrow U_A$, of dimension $d + 2d = b$. Elements of A annihilate D_A , and the annihilator has dimension $3d - 2d = d$, so it equals A . For the coordinate lines Fe_i , Q vanishes on Fe_i , so U_{Fe_i} contains $Fe_i \oplus 0$ and $0 \oplus D_{Fe_i}$; these span $X \oplus B$. $\square$

Lemma 3.3 (One update; Lemma 3.3 of [1]). *For uniform (x, y) and an independent uniform nonzero $\zeta \in U_A$, $\Pr[C_{\text{blk}}((x, y) + \zeta) \neq C_{\text{blk}}(x, y)] = 1 - \theta$.*

Proof. For $\zeta = (a, Q(a) + w)$ the output changes by $Q(a) + w + Q(x + a) + Q(x) = w + D(x, a)$, which lies in D_A . If ζ is uniform in all of U_A , then w is uniform in D_A and independent of a , so the change is uniform in D_A and is zero with probability q^{-2} . The zero update has probability q^{-3} , so conditioning on $\zeta \neq 0$ gives change probability $(1 - q^{-2})/(1 - q^{-3}) = q(q + 1)/(q^2 + q + 1) = 1 - \theta$. $\square$

So each block absorbs a random update with the small probability θ . That is the “one life” rate.

Linear observers of one block. A functional on $X \oplus B$ whose restriction to the vertical B is $z \in B^*$ has the form (γ_z, z) with $\gamma_z \in X^*$. Fix a subspace $S \leq B^*$ and a linear choice $\gamma: S \rightarrow X^*$ of extensions; write $\gamma_z(x) = \text{Tr}(g(z) \cdot x)$ with $g: S \rightarrow F^3$ binary linear. Restricting to an update space gives

$$\kappa_{A, \gamma}: S \rightarrow U_A^*, \quad z \mapsto (\gamma_z, z)|_{U_A}.$$

An observer “loses a dimension” at A when $\kappa_{A, \gamma}$ has a kernel.

Lemma 3.4 (When an observer dies; Lemma 3.4 of [1]). *For nonzero $z \in S$: $\kappa_{A, \gamma}(z) = 0$ if and only if $A = Fz$ and*

$$z \cdot g(z) = \sqrt{z_1 z_2 z_3}. \quad (2)$$

Proof. Vanishing on the vertical vectors $(0, w)$, $w \in D_A$, forces $z \in A$ by Lemma 3.2, so $A = Fz$. On the remaining vectors $(tz, Q(tz))$ the functional equals $\text{Tr}(t g(z) \cdot z + z \cdot Q(tz))$, and $z \cdot Q(tz) = 3t^2 z_1 z_2 z_3 = t^2 z_1 z_2 z_3$. Since $\text{Tr}(t^2 c) = \text{Tr}(t \sqrt{c})$, this is $\text{Tr}(t(g(z) \cdot z + \sqrt{z_1 z_2 z_3}))$, which vanishes for all t exactly when (2) holds. $\square$

So two things must go wrong for an observer to lose a dimension: the random line must be the line through one of its characters, and that character must satisfy the “alignment” equation (2). The left side of (2) is quadratic in the binary coordinates of z , while the right side is genuinely cubic, so typical subspaces have few aligned characters.

3.4 Generic subspaces

Definition 3.5. A binary subspace $S \leq F^3$ of dimension $r \geq 1$ is *generic* if no two distinct nonzero elements of S are F -proportional, and for every binary linear $g: S \rightarrow F^3$ at most $3r$ nonzero $z \in S$ satisfy (2).

For a generic S and a *fixed* γ , a uniformly random line A causes a loss only if $A = Fz$ for an aligned z , and at most one dimension is lost (two kernel elements would lie on the same line). Hence, writing r_A for the rank after restriction,

$$r - r_A \leq 1, \quad \Pr_A[r_A < r] \leq 3r\theta, \quad \mathbb{E}_A[H_r - H_{r_A}] \leq 3\theta. \quad (3)$$

For an arbitrary rank- r space a loss still needs $A = Fz$ for some nonzero $z \in S$, so $\Pr_A[r_A < r] \leq 2^r\theta$.

Lemma 3.6 (Lemma 3.6 of [1]). *For fixed r , a uniformly random r -dimensional binary subspace of F^3 is generic with probability tending to 1 as $d \rightarrow \infty$.*

Proof sketch. Parametrize S by independent uniform $x_{ij} \in F$ through $z_i(t) = \sum_j x_{ij}t_j$ for $t \in \mathbb{F}_2^r$. Non-injectivity and proportional pairs have probability $O_r(q^{-2})$.

For alignment, fix a set E of $3r + 1$ nonzero points t . Since g is determined by its values g_{ij} on a basis, alignment on all of E says that the vector $c = (\sqrt{z_1(t)z_2(t)z_3(t)})_{t \in E}$ lies in the F -span of the $3r$ columns $M_{t,(i,j)} = t_j z_i(t)$. Treat the x_{ij} as indeterminates over $L = \mathbb{F}_2(x_{ij})$. Because the square root is additive in characteristic 2 and $t_j \in \{0, 1\}$,

$$c_t = \sum_{j,k,l} t_j t_k t_l \sqrt{x_{1j}x_{2k}x_{3l}},$$

and the r^3 elements $\sqrt{x_{1j}x_{2k}x_{3l}}$ are linearly independent over L (distinct parity patterns of exponents). If c were in the column span over $\bar{L}$, every L -row vector killing the columns would kill each evaluation vector $(t_j t_k t_l)_{t \in E}$, so all of these, which include all monomials of degree 1, 2, 3 (as $t_i^2 = t_i$), would lie in the L -column span. But the columns are evaluations of polynomials of degree ≤ 2 . Then “degree ≤ 3 ” $\subseteq$ column span $\subseteq$ “degree ≤ 2 ”, so these spaces coincide and are closed under multiplication by each t_i , hence contain all positive-degree monomials, which span L^E because $0 \notin E$. That forces the column span to have dimension $|E| > 3r$, a contradiction.

To specialize to F , square everything (Frobenius preserves rank): the augmented matrix then has a nonzero minor of size $\text{rank} + 1$ that is a polynomial of degree $\leq 6r + 3$, while all minors of that size of the unaugmented matrix vanish identically. By the Schwartz–Zippel bound, column membership after specialization has probability $\leq (6r + 3)/q$. A union bound over the finitely many E finishes the proof. $\square$

Fix r_0 (to be chosen from p_*), then $\varepsilon_0 \in (0, 1)$ with $H_{r_0} 2^{r_0} \varepsilon_0 \leq 1$, and then d so large that $3d \geq r_0$ and, for every $1 \leq r \leq r_0$, a uniform rank- r subspace of B^* is non-generic with probability at most ε_0 . Fix one generic subspace S_* of rank r_0 .

3.5 The tree of blocks

We now stack blocks, keeping the output space B fixed. We build input spaces P_t , “shift” subspaces $R_t \leq P_t$ with surjections $\lambda_t: R_t \rightarrow B$, and maps $C_t: P_t \rightarrow B$ satisfying

$$C_t(u + h) = C_t(u) + \lambda_t(h) \quad (h \in R_t). \quad (4)$$

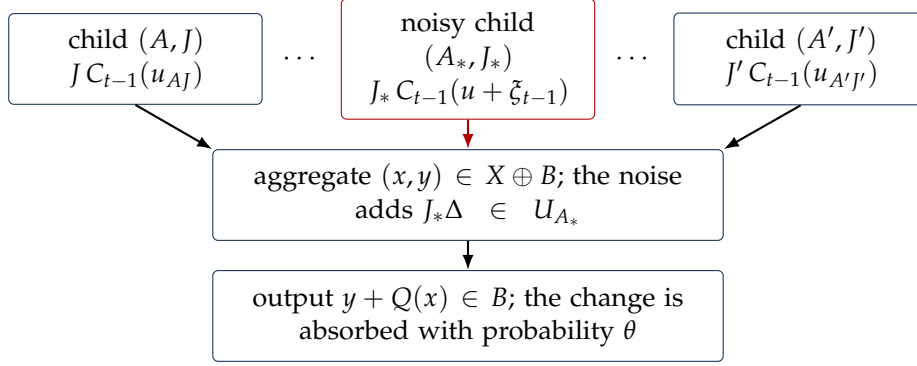


Figure 2: One level of the tree. Every child output is embedded into an update space U_A , the embedded outputs are summed, and one quadratic block is applied. Only the randomly selected child carries noise.

Start with $P_0 = R_0 = B$ and $C_0 = \lambda_0 = \text{id}$. Let $\mathcal{I}$ be the set of pairs (A, J) , with A a line and $J: B \rightarrow U_A$ a binary linear isomorphism. Put $P_t = \prod_{(A, J) \in \mathcal{I}} P_{t-1}$: one child for *every* line and every way of identifying the child's output with that line's update space. Then aggregate and apply one block:

$$(x, y) = \sum_{(A, J)} J C_{t-1}(u_{AJ}) \in X \oplus B, \quad C_t(u) = y + Q(x).$$

A tuple of child shifts is a parent shift when the X -part of its aggregate vanishes, and then λ_t records the B -part. Because the U_A span $X \oplus B$, λ_t is onto, and C_t maps uniform inputs to uniform outputs.

The noise. ζ_0 is a uniform element of B . For $t \geq 1$, ζ_t picks a uniform child (A, J) and puts an independent copy of ζ_{t-1} there, with zero elsewhere. Equivalently: choose a uniform root-to-leaf path, one uniform (A, J) per level, and add a uniform symbol of B at that leaf.

Lemma 3.7 (Nonlinear stability; Lemma 3.7 of [1]). *For uniform $u \in P_t$, $p_t := \Pr[C_t(u + \zeta_t) \neq C_t(u)] = (1 - q^{-3})(1 - \theta)^t$.*

Proof. At height 0 this is the probability that a uniform symbol is nonzero. Condition on the selected child and its output change Δ . By (4) the change is invariant under shifting that child's input, while its initial output is translated by every element of B , so the child's initial output is uniform given Δ ; all other child outputs are independent and uniform. Since the aggregate map is onto, the initial aggregate (x, y) is uniform and independent of the update $J\Delta$. Given $\Delta \neq 0$ and A , the isomorphism J is still uniform, so $J\Delta$ is uniform in $U_A \setminus \{0\}$, and Lemma 3.3 gives change probability $1 - \theta$. Hence $p_t = (1 - \theta)p_{t-1}$. $\square$

3.6 Linear observers of the tree

To track linear observers we need to say what an observer “restricting to z on the output” means on a tree whose output translations are realized by many different shifts.

Definition 3.8. For $S \leq B^*$, a *lift* on P_t is a linear map $z \mapsto \psi_z$ from S to P_t^* with $\psi_z|_{R_t} = z \circ \lambda_t$. It is *full* if $S = B^*$.

Lemma 3.9 (Factorization; Lemma 3.8 of [1]). *Given a lift on P_t ($t \geq 1$), there is a single linear $\gamma: S \rightarrow X^*$ such that for every child (A, J) the restriction of ψ_z to that child's shifts factors through λ_{t-1} via the character*

$$\phi_{AJ}(z) = J^*((\gamma_z, z)|_{U_A}) \in B^*.$$

Proof idea. A shift supported on one child with $\lambda_{t-1} = 0$ is a parent shift of output 0, so ψ_z kills it; hence ψ_z restricted to child shifts is a character of the child outputs. Summing these characters over children and subtracting z applied to the B -part of the aggregate gives something that vanishes whenever the X -part of the aggregate vanishes, so it factors through the X -part: that is γ_z . $\square$

So descending into child (A, J) , the observer becomes $J^*\kappa_{A, \gamma}$, with the same γ for all children. Since J^* is an isomorphism, the rank after descending is the rank of $\kappa_{A, \gamma}$, and Lemma 3.4 and Equation (3) apply. Choose a linear section of ϕ_{AJ} by a fixed rule to get a lift on the child with domain $\text{im } \phi_{AJ}$; every functional obtained this way is a restriction of a member of the original family.

Proposition 3.10 (Detection at fixed B ; Prop. 3.9 of [1]). *Let $s = \lfloor H_{r_0}/(8\theta) \rfloor$. Every full lift on P_s detects ξ_s with probability at least $1/4$: some $\psi_z(\xi_s) \neq 0$.*

Proof. Restrict the lift to the generic space S_* of rank r_0 and follow the random leaf path, descending the lift as above. Let r_j be the rank after j steps and $\Delta_j = H_{r_j} - H_{r_{j+1}} \geq 0$.

Freshness. Let $\mathcal{F}_j$ be the path choices before step j . Given $\mathcal{F}_j$ and the next line A_j , the subspace $E_j = \text{im } \kappa_{A_j, \gamma_j} \leq U_{A_j}^*$ is fixed, and the next domain is $S_{j+1} = J_j^* E_j$. Because J_j is a uniform isomorphism, S_{j+1} is a *uniformly random* subspace of B^* of that rank, so it is non-generic with probability at most ε_0 . The next lift may depend on J_j , but genericity controls every g at once, so this does no harm.

Potential. Given $\mathcal{F}_j$, the line A_j is uniform. If S_j is generic, $\mathbb{E}[\Delta_j | \mathcal{F}_j] \leq 3\theta$ by (3); if not, a loss has probability $\leq 2^{r_0}\theta$ and size $\leq H_{r_0}$. Averaging with freshness, $\mathbb{E}\Delta_j \leq (3 + H_{r_0}2^{r_0}\varepsilon_0)\theta \leq 4\theta$. Summing, $\mathbb{E}[H_{r_0} - H_{r_s}] \leq 4s\theta \leq H_{r_0}/2$. Extinction requires a drop of H_{r_0} , so by Markov $\Pr[r_s = 0] \leq 1/2$.

At the leaf, $P_0 = R_0 = B$ and the lift consists of the surviving characters themselves; an independent uniform symbol of B is detected by a rank- $r_s > 0$ space with probability $1 - 2^{-r_s} \geq 1/2$. The product is $1/4$. $\square$

3.7 From the tree to the gadget

Quotient. Put $V_0 = P_s / \ker \lambda_s$. By (4), C_s descends to $\bar{C}: V_0 \rightarrow B$, and $R_s / \ker \lambda_s \cong B$ becomes an honest subspace $B \subseteq V_0$ with $\bar{C}(x + b) = \bar{C}(x) + b$. Let ν be the image of the noise. Stability is unchanged, and pulling back any $G_0 \leq V_0^*$ with $G_0|_B = B^*$ along a section gives a full lift, so G_0 detects ν with probability $\geq 1/4$.

Enlargement. Fix $\ell \geq b + 2$, $K = \mathbb{F}_2^\ell$, and let $\mathcal{L}$ be the set of all injective linear maps $L: B \rightarrow K$. Take one copy of V_0 for each L , put $\tilde{C}((x_L)_L) = \sum_L L \tilde{C}(x_L)$, and quotient by the shifts of total output 0; the result is $\mathcal{V}$ with $K \subseteq \mathcal{V}$ and $C: \mathcal{V} \rightarrow K$ equivariant. The noise μ picks a uniform L and adds a sample of v in that copy. Since L is injective, C changes exactly when that copy's output does.

Detection with a fixed threshold. Let $G \leq \mathcal{V}^*$ with $S = G|_K$ of dimension $r \geq r_* := b + 2$. On copy L , the restriction of G to that copy's B is L^*S . This fails to be all of B^* only if $Lb' \in S^\perp$ for some nonzero $b' \in B$; for fixed b' and uniform L , Lb' is a uniform nonzero vector, so this has probability $\leq 2^{-r}$, and a union bound gives $\Pr_L[L^*S \neq B^*] \leq 2^{b-r} \leq 1/4$. On the remaining copies detection is at least $1/4$, for a total of at least $\frac{3}{4} \cdot \frac{1}{4} \geq \frac{1}{8}$.

Proof of Theorem 3.1. Choose r_0 with $\exp(1 - H_{r_0}/8) \leq p_*$, then $\varepsilon_0, d, \theta, s$ as above. Stability: $p_s \leq (1 - \theta)^s \leq e^{-\theta s} \leq e^{\theta - H_{r_0}/8} \leq p_*$. Detection holds with $r_* = b + 2 = 3d + 2$, which depends only on p_* , for every $\ell \geq r_*$. All objects are finite and can be found by enumeration. $\square$

Remark 3.11 (Size). The gadget is enormous. $|\mathcal{I}|$ is about $q^2 |GL_{3d}(\mathbb{F}_2)| \approx 2^{9d^2}$, the tree has depth $s \approx q^2 \ln r_0/8$, so $\dim P_s$ is about $3d \cdot 2^{9d^2 s}$, doubly exponential in q , and $|\mathcal{V}|$ is exponential in that. This does not matter for the theorem, where $\mathcal{V}$ is a constant, but it makes the eventual polynomial running time astronomically large (Remark 10.1).

4 The Unique Games instance

Fix gadget data $(\mathcal{V}, K, C, \mu)$ from Theorem 3.1, with $K = \mathbb{F}_2^\ell$, and a linear complement $\mathcal{V} = K \oplus W$. Fix a tuple length k and the preprocessed parity instance $\mathcal{E}$ with weights ω .

4.1 Questions and answer spaces

An *equation question* is a tuple $U = (e_1, \dots, e_k)$ drawn from $\omega^{\otimes k}$, with replacement. Its answers are the k -tuples of satisfying assignments. Write an assignment to block i by its first two bits (x_i, y_i) ; the third is $b_{e_i} + x_i + y_i$. Add a homogeneous coordinate t_0 and set

$$L_U = \mathbb{F}_2^m, \quad m = 1 + 2k,$$

so that the *valid* answers are exactly the vectors with $t_0 = 1$. Write $e_0^\top$ for the functional reading t_0 . Repeated equations in different blocks get separate local coordinates.

A *projected question* O is obtained from U by independently, in each block, keeping the equation with probability $1 - \beta$, or with probability β keeping only one of its three variables, chosen uniformly. The projection $\tilde{\pi}: L_U \rightarrow L_O$ keeps t_0 , is the identity on kept equations, and sends a projected block to its retained bit:

$$x_i, \quad y_i, \quad \text{or} \quad b_{e_i} t_0 + x_i + y_i.$$

Note the t_0 term in the third case: the third variable is an *affine* function of the first two, which is why homogeneous coordinates are used.

4.2 Tables and exact keys

A *table* on a question Q (equation or projected) is a linear map $P \in \text{Hom}(L_Q, \mathcal{V})$, equivalently an affine map from valid answers to $\mathcal{V}$. Write it as $P = c t_0 + \sum_i (a_i x_i + d_i y_i)$ on an equation question, with intercept c and block slopes $(a_i, d_i) \in \mathcal{V}^2$.

The soundness proof will compare a tuple U with its projection O , and needs the labeling's answer on a table P' for O to coincide with its answer on the pulled-back table $P' \tilde{\pi}$ for U . Rather than test this, the construction makes the two tables literally the same vertex. The device is a canonical form, the *key*, which records only the coordinates a table actually depends on.

Reducing a block. In an equation block look at the rank of the slope map $(x_i, y_i) \mapsto a_i x_i + d_i y_i$:

- rank 2: keep the equation and its slopes;
- rank 0: drop the block;
- rank 1: the slopes are $(v, 0)$, $(0, v)$ or (v, v) for some $v \neq 0$, so the table depends on the block only through one actual variable, at position 1, 2 or 3. Record that variable and the slope v . In the third case $v(x_i + y_i) = v z_i + b_{e_i} v t_0$, where z_i is the third bit, so add $b_{e_i} v$ to the intercept.

In a singleton block of a projected question, keep the variable if its slope is nonzero and drop it otherwise.

Definition 4.1 (Key). The *support* of (Q, P) is the list of kept blocks, with their positions in $[k]$, their tags (equation or variable) and their IDs. The *key* is the support together with the reduced affine table on it, including the corrected intercept $\tilde{c} = c + \sum_{i \in J_3} b_{e_i} v_i$, where J_3 is the set of blocks reduced to their third variable.

Lemma 4.2 (Pullbacks share keys; §4.3 of [1]). *For every permitted projection $\tilde{\pi}: L_U \rightarrow L_O$ and every table P' on O , the tables $(U, P' \tilde{\pi})$ and (O, P') have the same key.*

Proof. Kept equations contribute identically. A singleton of slope 0 disappears on both sides. A singleton of nonzero slope v at position j pulls back to the rank-one pattern $(v, 0)$, $(0, v)$ or (v, v) , which reduces to the same variable at the same position. For position 3, the pullback adds $b_e v$ to the intercept (from the $b_e t_0$ term of $\tilde{\pi}$), and the reduction adds $b_e v$ again; over $\mathbb{F}_2$ the two cancel. $\square$

Example 4.3. Take $k = 1$, one equation $x + y + z = 1$, and a table on U with slopes (v, v) and intercept c , so $P(x, y) = c + v(x + y)$. On valid answers $z = 1 + x + y$, so $P = (c + v) + vz$: the key records “variable z , slope v , intercept $c + v$ ”. The projected question O that keeps only z , with table $P'(z) = (c + v) + vz$, pulls back to $(c + v) + v(1 + x + y) = c + v(x + y)$, the same table, with the same key.

4.3 Folding and vertices

Translating a key by $h \in K$ means adding h to its reduced intercept (realized by replacing P with $P + h e_0^\top$). This action is free, and the *vertices* of the instance are its orbits; each is represented by the key whose intercept has zero K -component. For a table P on Q let $v(Q, P)$

be the vertex and $c(Q, P) \in K$ the removed K -component. A labeling $\lambda: \text{vertices} \rightarrow K$ defines the *restored answer*

$$F_Q(P) = \lambda(v(Q, P)) + c(Q, P),$$

and by construction and [Lemma 4.2](#)

$$F_Q(P + h e_0^\top) = F_Q(P) + h \quad (h \in K), \quad F_U(P' \bar{\pi}) = F_O(P'). \quad (5)$$

These two identities, *folding* and *sharing*, are the only properties of the vertex construction the soundness proof uses.

4.4 The test, and completeness

The instance $\mathcal{G}$ samples, independently,

$$U \sim \omega^{\otimes k}, \quad P \in \text{Hom}(L_U, \mathcal{V}) \text{ uniform}, \quad a \sim \mu, \quad l \in \mathbb{F}_2^m \text{ uniform},$$

and checks

$$F_U(P) = F_U(P + a l^\top). \quad (6)$$

In terms of the labeling this reads $\lambda(v_1) = \lambda(v_0) + c_0 + c_1$, a translation constraint on K . The weights are the sampling probabilities. The projected questions O do not appear in the test; they exist only in the analysis.

Lemma 4.4 (Completeness; Lemma 4.3 of [1]). *If $\text{OPT}(\mathcal{E}) \geq 1 - \zeta$, then $\text{val}(\mathcal{G}) \geq 1 - k\zeta - p_*/2$.*

Proof. Fix a global assignment of value $\geq 1 - \zeta$. It gives an answer on every support: the global bit on a singleton; on an equation, its first two global bits and the third bit that satisfies the equation. Label each orbit representative by applying C to its reduced table evaluated at this answer. By equivariance of C , the restored answer of any table is then C of the table evaluated at the support answer.

With probability $\geq 1 - k\zeta$ every equation in U is satisfied, and then all support answers are projections of one full answer $z_U \in L_U$ with $t_0 = 1$. The two sides of (6) become $C(Pz_U)$ and $C(Pz_U + a(l^\top z_U))$. Since $z_U \neq 0$, Pz_U is uniform in $\mathcal{V}$ and independent of (a, l) , and $l^\top z_U$ is a fair coin, so the test fails with probability at most $p_*/2$. $\square$

The honest labeling is a function of the key only, which is why it is well-defined on vertices: two tables with the same key have the same support, hence the same support answer.

5 The plan for soundness

Soundness means: on a NO instance, no labeling passes (6) with probability 0.99. As in the 2-to-2 proof, this is shown by playing two bounds against each other in an auxiliary two-player game with *advice*.

5.1 Parameters

The constants are chosen in this order, each depending only on the earlier ones:

1. $p_* > 0$, and the gadget threshold r_* (Theorem 3.1);
2. $\eta = 2^{-r_*}/16$;
3. from the inverse shortcode theorem (Theorem 7.1), $\alpha \in (0, 1]$ and $r_s \geq 1$ depending on η ;
4. ℓ large enough for the inverse theorem and the gadget, with $2^{-(\ell-r_s)} < \alpha/8$ and $2^{-\ell} < \eta/4$; then the gadget $(\mathcal{V}, K, C, \mu)$ and a splitting $\mathcal{V} = K \oplus W$;
5. the tuple length k , a perfect cube, and $\beta = k^{-2/3}$.

Write a table as $P = (M, T)$ with $M: L_U \rightarrow K$ and $T: L_U \rightarrow W$. The K -part M is an $\ell \times m$ binary matrix, and the BKS shortcode lives on it.

5.2 The advice game

Definition 5.1 (Projected advice experiment; Def. 5.2 of [1]). Sample U, O and $\bar{\pi}$ as in Section 4; a uniform linear map $A: K \rightarrow \mathbb{F}_2^{r_s}$; and uniform tables $T_O: L_O \rightarrow W$ and $M_O: L_O \rightarrow K$. Put $S_O = AM_O$. The two players receive

$$U\text{-player: } (U, A, T_O \bar{\pi}, S_O \bar{\pi}), \quad O\text{-player: } (O, A, T_O, S_O),$$

and each returns a valid answer for its own question. They win if $\bar{\pi}(a_U) = a_O$.

Both players see the W -part of the table and r_s random linear combinations of the rows of the K -part, pulled back appropriately; neither sees the rest of M_O . The advice has bounded dimension per block, $\dim W + r_s$, independent of k , and that is what the upper bound exploits.

Why advice? The inverse theorem will say that a good labeling, restricted to a typical (U, T) , agrees with an affine function on a *slice* of matrices cut out by a few row conditions $d_i^\top M = s_i^\top$ and column conditions $Mq_j = t_j$. To turn that into an answer, a player must know which slice it is in. Column conditions are about answers (the q_j live in L_U), but row conditions are about the table. Giving both players the values of a few random row combinations AM lets them locate a row fiber, and Section 7.2 shows that a random fiber is useful with constant probability. This is the “advice” idea of Khot–Minzer–Safrá and Minzer–Zheng.

5.3 The two bounds

Proposition 5.2 (Decoding lower bound; Prop. 5.3 of [1]). *There are $\gamma > 0$ and k_0 , depending only on the fixed data of items 1–4, such that for every cube $k \geq k_0$ and every parity instance: if a family of functions $F_Q: \text{Hom}(L_Q, \mathcal{V}) \rightarrow K$ satisfies folding and sharing (5) and passes (6) with probability ≥ 0.99 , then the advice experiment with $\beta = k^{-2/3}$ has strategies that win with probability $\geq \gamma$.*

Lemma 5.3 (Repetition upper bound; Lemma 6.2 of [1]). *If $\text{OPT}(\mathcal{E}) \leq 4/5$, every pair of strategies in the advice experiment wins with probability at most*

$$\left(1 - \frac{\beta 2^{-\dim W - r_s}}{3600}\right)^k \leq \exp\left(-\frac{2^{-\dim W - r_s}}{3600} k^{1/3}\right).$$

Proposition 5.4 (A fixed gap; Prop. 6.3 of [1]). *For every sufficiently large cube k ,*

$$\text{OPT}(\mathcal{E}) \geq 1 - \xi \Rightarrow \text{val}(\mathcal{G}) \geq 1 - k\xi - p_*/2, \quad \text{OPT}(\mathcal{E}) \leq 4/5 \Rightarrow \text{val}(\mathcal{G}) \leq 99/100.$$

The threshold on k depends only on the fixed data, not on $\mathcal{E}$ or ξ .

Proof. Completeness is Lemma 4.4. Every labeling yields a family with folding and sharing. If it passed with probability ≥ 0.99 , Proposition 5.2 would give winning probability $\geq \gamma$, but for large k the bound of Lemma 5.3 is below γ . $\square$

With Corollary 2.3, Proposition 5.4 proves Theorem 1.1; the bookkeeping is in Section 10. Note that the NO bound 99/100 does not depend on p_* : the gadget's completeness can be made as good as desired without touching soundness.

The proof of Proposition 5.2 has three steps:

1. **Spectral comparison** (Section 6): high acceptance of the latent test implies acceptance $\geq 0.9 \cdot 2^{-r_*}$ of the ordinary BKS test on the K -part, for typical (U, T) .
2. **Structure and abundance** (Section 7): by the inverse shortcode theorem and an erasure argument, a random row fiber is “good” with constant probability, and each good fiber determines a short list of valid answers.
3. **Transfer and decoding** (Section 8): the structure survives the passage from U to its projection O , and a Fourier sampling rule for the O -player hits the projection of the U -player's list.

6 From the latent test to the ordinary shortcode test

This is the step where the gadget is used, and it is short.

Lemma 6.1 (Spectral comparison; Lemma 5.4 of [1]). *If the family passes (6) with probability ≥ 0.99 , then*

$$\mathbb{E}_{U,T} \Pr_{M,a,l} [F_U(M, T) = F_U(M + al^\top, T)] \geq 0.9 \cdot 2^{-r_*},$$

where now a is uniform in K , and M, T, l are uniform. Consequently at least a 10η fraction of pairs (U, T) have inner probability $\geq 4\eta$.

Proof. Fix U . For a character $\sigma \in K^*$ let $f_\sigma(P) = (-1)^{\sigma(F_U(P))}$; then $\Pr[F_U(P) = F_U(P')] = \mathbb{E}_\sigma \mathbb{E}[f_\sigma(P)f_\sigma(P')]$ for uniform σ . Fourier-expand f_σ on $\text{Hom}(L_U, \mathcal{V}) \cong \mathcal{V}^m$: the characters are indexed by $g = (g_0, \dots, g_{m-1}) \in (\mathcal{V}^*)^m$, with $\chi_g(P) = (-1)^{\sum_i g_i(Pe_i)}$. Adding $al^\top$ multiplies χ_g by $(-1)^{\sum_i l_i g_i(a)}$. Averaging over uniform l gives 1 if all $g_i(a) = 0$ and 0 otherwise, so for any noise law the acceptance is

$$\mathbb{E}_{U,\sigma} \sum_g |\hat{f}_\sigma(g)|^2 \Lambda_g, \quad \Lambda_g = \Pr_a [g_i(a) = 0 \forall i] \in [0, 1].$$

The multipliers are nonnegative, which is what makes the comparison work.

Latent noise. For $a \sim \mu$, the gadget's detection property applied to $G = \text{span}\{g_i\}$ gives $\Lambda_g \leq 7/8$ whenever $\text{rk}(g) := \dim \text{span}\{g_i|_K\} \geq r_*$. If B is the average Fourier mass on such “high-rank” g , then $0.99 \leq 1 - B/8$, so $B \leq 0.08$: at least 0.92 of the mass is on $\text{rk}(g) < r_*$.

Uniform noise in K . For a uniform in K , $\Lambda_g = 2^{-\text{rk}(g)} \geq 2^{-(r_*-1)}$ on low-rank g . Dropping the (nonnegative) high-rank terms, acceptance is at least $0.92 \cdot 2^{1-r_*} \geq 0.9 \cdot 2^{-r_*}$. A perturbation in K changes only the K -part M and leaves T fixed, so this is the average over (U, T) of the ordinary BKS equality test applied to $M \mapsto F_U(M, T)$.

The average is $0.9 \cdot 2^{-r_*} = 14.4\eta$. If a q fraction of pairs have inner probability $\geq 4\eta$, then $14.4\eta \leq 4\eta + q$, so $q \geq 10\eta$. $\square$

What just happened. The gadget cannot make dishonest labelings fail outright; it only damps their high-rank Fourier mass by a constant factor $7/8$. That is enough because the target soundness is a fixed 0.99 , not a small δ . Since r_* was fixed before ℓ , the conclusion is an equality-test acceptance of order 2^{-r_*} , a constant that does not shrink as the alphabet grows, which is exactly what the KMS inverse theorem needs.

7 Structure: the inverse theorem and good advice

7.1 The inverse shortcode theorem

A *slice* of $\text{Mat}_{\ell, m}$ is a nonempty set

$$\mathcal{S} = \{M : d_i^\top M = s_i^\top \ (i \leq n_r), \ Mq_j = t_j \ (j \leq n_c)\},$$

cut out by n_r row conditions and n_c column conditions.

Theorem 7.1 (Inverse shortcode theorem; Thm. 5.1 of [1]). *For every $\eta \in (0, 1)$ there are $\alpha \in (0, 1]$, $r_s \geq 1$ and ℓ_0 such that for all $\ell \geq \ell_0$ and all $m \geq m_0(\eta, \ell)$: if $f : \text{Mat}_{\ell, m} \rightarrow \mathbb{F}_2^\ell$ satisfies $\Pr_{M, a, l}[f(M) = f(M + al^\top)] \geq \eta$, then there are a slice $\mathcal{S}$ with $n_r, n_c \leq r_s$ and $z \in \mathbb{F}_2^m$, $u \in \mathbb{F}_2^\ell$ with $\Pr_{M \in \mathcal{S}}[f(M) = Mz + u] \geq \alpha$.*

Derivation from Grassmann expansion. Equality acceptance is the mass-weighted average of how well each level set $f^{-1}(y)$ is preserved by a random rank-one step, so some level set S is preserved with probability $\geq \eta$. Map each matrix to the ℓ -dimensional subspace $L_M = \{(x, M^\top x)\} \subseteq \mathbb{F}_2^{\ell+m}$. Then $\dim(L_M \cap L_N) = \ell - \text{rank}(M - N)$, so rank-one steps are exactly Grassmann edges inside this chart, and exactly half of each vertex's Grassmann neighbors lie in the chart. Hence $\{L_M : M \in S\}$ has Grassmann edge retention at least about $\eta/4$. The Khot–Minzer–Safra theorem [11] says such a set has density $\geq \alpha(\eta)$ inside some “zoom-in/zoom-out” interval $\{L : A_0 \subseteq L \subseteq B_0\}$ with $\dim A_0 + \text{codim } B_0 \leq r_s$. In matrix coordinates, $A_0 \subseteq L_M$ is a set of row conditions and $L_M \subseteq B_0$ a set of column conditions. On that slice f equals the constant y on an α fraction, which is the claim with $z = 0$. $\square$

The proof gives $z = 0$. Nonzero targets z appear only after the column conditions are used to shift z (Lemma 7.4).

7.2 Abundance of good advice

The inverse theorem gives one good slice for each good (U, T) . But the U -player sees only a *random* row fiber $\{M : AM = AM_{\text{actual}}\}$, and a specific fiber is hit with probability about

$2^{-r_s m}$, which vanishes as k grows. The next lemma shows that good fibers are, in aggregate, abundant.

Definition 7.2. Row advice (U, T, A, S_0) is *good* if there are at most r_s column conditions and a target $Mz + u$ such that the set of M satisfying both $AM = S_0$ and the column conditions is nonempty and $F_U(M, T) = Mz + u$ on at least an $\alpha/2$ fraction of it.

Lemma 7.3 (Good advice is common; Lemma 5.5 of [1]). *For large k , if the family passes (6) with probability ≥ 0.99 , then*

$$\Pr_{U, T, A, M} [(U, T, A, AM) \text{ is good}] \geq g_0 := 10\eta^2 2^{-\ell r_s}.$$

Proof. Fix (U, T) with inner acceptance $\geq 4\eta$ (Lemma 6.1) and write $f(M) = F_U(M, T)$. Let B be the union of all good row fibers, over all A and S_0 . Suppose $\Pr[B] < \eta$.

Erase. Replace f on B by independent uniform values. Both endpoints of a test step are uniform matrices, so acceptance drops by at most $2\Pr[B]$, leaving at least 2η .

The erased values match nothing. A slice with at most r_s row and r_s column conditions has at least $2^{(\ell-r_s)(m-r_s)}$ elements, and there are at most $2^{(2r_s+1)(\ell+m)}$ slice-and-target descriptions. For each, the expected number of random matches is at most $2^{-\ell}|\mathcal{S}| < \alpha|\mathcal{S}|/8$, and Hoeffding plus a union bound show that, for large m , some choice of erased values contributes fewer than $\alpha|\mathcal{S}|/4$ matches to *every* slice and target.

Contradiction. Take any slice of the allowed type and any target. Its row conditions define a row fiber for some A (pad with zero rows). If that fiber was good, all of it was erased, and agreement is $< \alpha/4$. If it was not good, the original agreement was $< \alpha/2$, and erasure adds $< \alpha/4$. Either way agreement is $< \alpha$, contradicting Theorem 7.1 for the modified function, whose acceptance is $\geq 2\eta$. (The inverse theorem does not need folding, so erasure is harmless.)

So $\Pr[B] \geq \eta$. Each $M \in B$ lies in a good fiber for some particular A , which a uniform A hits with probability $2^{-\ell r_s}$. Averaging over the $\geq 10\eta$ good pairs (U, T) gives the bound. $\square$

7.3 Folding turns a slice into answers

Lemma 7.4 (The answer set; Lemma 5.5 of [1], second part). *For good advice, fix (by a deterministic rule depending only on the advice and the family) a witness: column conditions $Mq_j = t_j$ and a target $Mz + u$. Let $Z = \text{span}\{q_j\}$. Then z can be chosen with $e_0^\top z = 1$, and*

$$\mathcal{A} = (z + Z) \cap \{v : e_0^\top v = 1\}$$

is a nonempty set of valid answers of size at most 2^{r_s} .

Proof. On the slice Mw is fixed for $w \in Z$, so replacing z by $z + w$ and u by $u + Mw$ does not change the target. Suppose every vector of $z + Z$ had $t_0 = 0$. For $h \in H = \ker A$, the shift $M \mapsto M + h e_0^\top$ preserves the row fiber (as $Ah = 0$), the column conditions (as $e_0^\top q_j = 0$) and the target (as $e_0^\top z = 0$), while folding shifts F_U by h . Each orbit of this free action contains at most one agreeing matrix, so agreement is at most $|H|^{-1} \leq 2^{-(\ell-r_s)} < \alpha/8$, contradicting goodness. $\square$

Folding is what makes answers appear. The inverse theorem by itself only says “ f is constant on a slice”. Folding forbids f from being constant along the direction $e_0^\top$, so the slice must pin down the table along some answer-like vector with $t_0 = 1$. Those vectors are the decoded answers. This is the matrix version of the standard fact that a folded long code cannot be biased toward the empty set.

8 Transfer and Fourier decoding

So far the U -side tables were uniform. In the advice experiment the U -player’s table is a padded table $(M_O \bar{\pi}, T_O \bar{\pi})$ from O . The choice $\beta = k^{-2/3}$ makes these almost the same.

8.1 Padded tables look uniform

Lemma 8.1 (Lemma 5.6 of [1]). *For every U , averaging over the projection choices, the law of $(M_O \bar{\pi}, T_O \bar{\pi})$ is within total variation $o(1)$ of the uniform law on $\text{Hom}(L_U, \mathcal{V})$ as $k \rightarrow \infty$ along cubes, uniformly in U and the instance.*

Proof. Let $n = |\mathcal{V}|$. In a kept block the two slopes are independent uniform elements of $\mathcal{V}$ (law P_0). In a projected block the pulled-back slopes are $(a, 0)$, $(0, a)$ or (a, a) with a uniform, each with probability $1/3$ (law Q_0). The likelihood ratio dQ_0/dP_0 is n at $(0, 0)$, $n/3$ at the other $3(n-1)$ points of the three lines, and 0 elsewhere, so $\chi^2(Q_0 \| P_0) = (n-1)/3$. The block mixture $(1-\beta)P_0 + \beta Q_0$ has $\chi^2 = \beta^2(n-1)/3$ from P_0 , and over k independent blocks

$$\chi^2 = \left(1 + \frac{n-1}{3}\beta^2\right)^k - 1 \rightarrow 0, \quad \text{since } k\beta^2 = k^{-1/3} \rightarrow 0.$$

Total variation is at most $\frac{1}{2}\sqrt{\chi^2}$. The intercept stays uniform because the extra terms $b_e a$ from third-position projections are added to an independent uniform intercept. $\square$

The two jobs of $\beta = k^{-2/3}$. $k\beta^2 \rightarrow 0$ makes padded tables indistinguishable from uniform ones, so everything proved for uniform tables transfers. $k\beta \rightarrow \infty$ makes the number of projected blocks grow, which the repetition bound in Section 9 needs. Any β with $k^{-1} \ll \beta \ll k^{-1/2}$ would do.

8.2 The witness survives projection

A column condition $Mq = t$ on U becomes $M_O(\bar{\pi}q) = t$ on O . This can be inconsistent (if $\bar{\pi}q = 0 \neq t$), so survival must be argued.

Lemma 8.2 (Transfer; Lemma 5.7 of [1]). *For large cubes k , with probability at least $\gamma_1 = \frac{\alpha g_0}{8} 2^{-\ell r_s}$ over the joint data $D = (U, O, \bar{\pi}, A, T_O, S_O)$, the U -player’s advice is good, and its witness transfers to a nonempty slice $\{M_O : AM_O = S_O, M_O \bar{\pi} q_j = t_j\}$ on which $F_O(M_O, T_O) = M_O \bar{\pi} z + u$ with probability at least $\alpha/4$.*

Proof. Under uniform sampling on U , let $\mathcal{J}$ be “the advice is good and M satisfies its column conditions” and $\mathcal{M}$ “ $F_U(M, T)$ hits the target”. A good fiber’s column slice costs at most ℓ_{r_s} extra linear conditions, so $\Pr[\mathcal{J}] \geq g_0 2^{-\ell_{r_s}}$ and $\Pr[\mathcal{J} \cap \mathcal{M}] \geq \frac{\alpha}{2} \Pr[\mathcal{J}]$, giving

$$\Pr[\mathcal{J} \cap \mathcal{M}] - \frac{\alpha}{4} \Pr[\mathcal{J}] \geq \frac{\alpha g_0}{4} 2^{-\ell_{r_s}}.$$

Both events depend only on (U, A, M, T) , so by [Lemma 8.1](#) the same holds, up to $o(1)$, with the padded table; for large k the difference is at least γ_1 . Now condition on D : the remaining M_O is uniform on its row fiber, and sharing gives $F_U(M_O \bar{\pi}, T_O \bar{\pi}) = F_O(M_O, T_O)$. The padded difference becomes $\mathbb{E}_D[p_D(q_D - \alpha/4)]$, where p_D is the probability of the transferred slice and q_D the agreement on it. Since the integrand is at most 1 and is nonpositive unless $p_D > 0$ and $q_D \geq \alpha/4$, that event has probability $\geq \gamma_1$. $\square$

8.3 The two decoders

The U -player. From its advice it computes (by the fixed rule) the witness and answer set $\mathcal{A}$ of [Lemma 7.4](#), and outputs a uniform element of $\mathcal{A}$; if the advice is not good it outputs any valid answer.

The O -player. It never sees the U -player’s witness. Let $H = \ker A \leq K$. Choose a representative M_0 with $AM_0 = S_O$, so the row fiber is $M_0 + X$ with $X = \text{Hom}(L_O, H)$. Pick a uniform $\sigma \in K^*$, form $f_\sigma(N) = (-1)^{\sigma(F_O(M_0+N, T_O))}$ on X , and sample a frequency $\Phi \in X^* = H^* \otimes L_O$ with probability $|\widehat{f_\sigma}(\Phi)|^2$. If $\sigma|_H \neq 0$, pick $h \in H$ with $\sigma(h) = 1$ and output the contraction $\Phi(h) \in L_O$.

The output is always valid: folding gives $f_\sigma(N + h e_0^\top) = -f_\sigma(N)$ when $\sigma(h) = 1$, so every Φ with nonzero coefficient has $\langle \Phi, h e_0^\top \rangle = 1$, that is, $e_0^\top \Phi(h) = 1$. This is Fourier list decoding in the style of Håstad and Khot–Safra, run on the row fiber the advice reveals.

Lemma 8.3 (Agreement; Lemma 5.8 of [1]). *On the data of [Lemma 8.2](#), the two decoders agree with probability at least $2^{-\ell_{r_s}}(\alpha/8)^2 2^{-r_s}$.*

Proof. Let $z' = \bar{\pi}z$ and $Z' = \bar{\pi}Z$, so $\bar{\pi}\mathcal{A} = (z' + Z') \cap \{t_0 = 1\}$. On the transferred slice $N|_{Z'}$ is a fixed map b . Define $C_\sigma = \mathbb{E}[f_\sigma(N)(-1)^{\sigma(Nz')} \mid N|_{Z'} = b]$. Expanding the agreement $\geq \alpha/4$ in characters of K gives $\mathbb{E}_\sigma[|C_\sigma|] \geq \alpha/4$; discarding the $\leq 2^{-(\ell-r_s)} < \alpha/8$ fraction of σ trivial on H leaves $\mathbb{E}_\sigma[\mathbf{1}_{\sigma|_H \neq 0} |C_\sigma|] \geq \alpha/8$.

Since $(-1)^{\sigma(Nz')}$ is the character $\theta_\sigma = \sigma|_H \otimes z'$ and the slice is a coset of $\{N : N|_{Z'} = 0\}$, whose annihilator is $H^* \otimes Z'$, C_σ is a signed sum of $\widehat{f_\sigma}(\Phi)$ over $\Phi \in \theta_\sigma + H^* \otimes Z'$, a coset of size at most $2^{\ell_{r_s}}$. Cauchy–Schwarz twice gives

$$\mathbb{E}_\sigma \left[\mathbf{1}_{\sigma|_H \neq 0} \sum_{\Phi \in \theta_\sigma + H^* \otimes Z'} |\widehat{f_\sigma}(\Phi)|^2 \right] \geq 2^{-\ell_{r_s}} (\alpha/8)^2.$$

Any such $\Phi = \sigma|_H \otimes z' + \Psi$ with $\Psi \in H^* \otimes Z'$ has $\Phi(h) = z' + \Psi(h) \in z' + Z'$, and it is valid, so $\Phi(h) \in \bar{\pi}\mathcal{A}$. The U -player independently picks a preimage with probability $\geq 1/|\mathcal{A}| \geq 2^{-r_s}$. $\square$

Proof of [Proposition 5.2](#). Combine [Lemmas 8.2](#) and [8.3](#): $\gamma = \gamma_1 2^{-\ell_{r_s}} (\alpha/8)^2 2^{-r_s}$, independent of k and of the instance. The strategies need not be efficient; they only bound the value of a finite game. $\square$

9 Clean coordinates and the repetition bound

It remains to prove [Lemma 5.3](#): on a NO instance every pair of strategies in the advice game wins with probability $\exp(-\Omega(k^{1/3}))$. The idea, from Khot–Minzer–Safra, is that on blocks where the advice happens to carry no information, the players are just playing $G_{\mathcal{E}}$ in parallel.

Proof of Lemma 5.3. Fix strategies. Condition on A , and combine the advice into $D_O = (T_O, S_O): L_O \rightarrow E_A := W \oplus \text{im } A$. Given the questions, the intercept and every block slope of D_O are independent and uniform in E_A , whatever the equation IDs and positions are.

Clean blocks. Call block i *clean* if it was projected to a variable and its advice slope is 0. Given A , blocks are clean independently with probability $p_A = \beta/|E_A| = \beta 2^{-\dim W - \text{rank } A}$, and

$$\Pr[e_i = e, i \text{ clean at position } j \mid A] = \omega_e \cdot \frac{1}{3} \cdot p_A,$$

so a clean block’s (equation, variable) pair has exactly the distribution of $G_{\mathcal{E}}$.

Freeze everything else. Let R consist of A , the clean set I , the intercept of D_O , and, for each $i \notin I$, its equation, whether it was projected, its position, and its slopes. Given $R = r$, the clean pairs $(e_i, v_{e_i, j_i})_{i \in I}$ are independent samples of $G_{\mathcal{E}}$.

Each player can rebuild its own input. The U -player, given the clean equations $(e_i)_{i \in I}$ and r , knows all of U . Its pulled-back slopes in clean blocks are $(0, 0)$; all other slopes and the full intercept $c_U = c_O + \sum_{j_i=3} b_{e_i} d_i$ are determined by r , because clean blocks contribute $d_i = 0$. So it can reconstruct its entire advice. Symmetrically the O -player, given the clean variables and r , reconstructs O and its advice. Neither needs the other’s clean questions.

Reduce to repetition. So for each r we get strategies for $G_{\mathcal{E}}^{\otimes |I|}$: rebuild the original input, run the original strategy, and report the answers in the clean blocks. Winning the advice game implies winning every clean block, so by [Theorem 2.1](#) and (1) (with $g = 1/15$, $g^2/16 = 1/3600$)

$$\Pr[\text{win} \mid R = r] \leq \text{val}(G_{\mathcal{E}}^{\otimes |I|}) \leq (1 - 1/3600)^{|I|}.$$

Given A , $|I| \sim \text{Bin}(k, p_A)$, so averaging gives $\mathbb{E}_A(1 - p_A/3600)^k \leq (1 - \beta 2^{-\dim W - r_s}/3600)^k$, using $\text{rank } A \leq r_s$, and $k\beta = k^{1/3}$. $\square$

Why the advice does not help a cheater. The advice has $\dim W + r_s$ bits per block, a constant. So a constant fraction of projected blocks get zero advice, and with $k\beta = k^{1/3}$ projected blocks there are $\Theta(k^{1/3})$ clean ones. On those, the advice is literally zero on both sides, so it cannot correlate the players beyond what the base game allows. Everything else is frozen and becomes shared randomness, which never helps in a two-player game.

10 Putting it together

10.1 The order of the parameters

The proof is a long chain of “for every ... there is ...”, and nearly every subtlety is about ordering. [Table 1](#) lists the choices in the order they are made. Each depends only on what is above it; in particular nothing depends on the input formula.

Parameter	Chosen so that
1 ε, δ	given
2 repetitions t	$(1 - 1/10,240,000)^t \leq \delta$ (depends on δ only)
3 p_*, τ	$p_* \leq \varepsilon/t, \tau \leq \min\{1/200, \varepsilon/t\}$
4 $r_0, \varepsilon_0, d, s; r_* = 3d + 2$	gadget stability $\leq p_*$ (Theorem 3.1)
5 $\eta = 2^{-r_*}/16; \alpha, r_s$	inverse shortcode theorem at η
6 ℓ ; then $(\mathcal{V}, K, C, \mu), W$	$\ell \geq \ell_0, 2^{-(\ell-r_s)} < \alpha/8, 2^{-\ell} < \eta/4$
7 g_0, γ_1, γ	explicit in η, α, r_s, ℓ
8 cube $k, \beta = k^{-2/3}$	$m = 1 + 2k \geq m_0$; erasure union bound < 1 ; TV small; $e^{-ck^{1/3}} < \gamma$
9 ζ	$\zeta < \min\{1/100, \varepsilon/(2kt)\}$
10 input φ	Håstad(ζ) $\rightarrow$ distinct positions $\rightarrow \mathcal{G} \rightarrow \text{subdivide}(\tau) \rightarrow t\text{-fold product}$

Table 1: Parameter order. The NO-case bound 0.99 uses only rows 4–8 and holds for every p_* .

Proof of Theorem 1.1. With the choices of Table 1, Proposition 5.4 gives a weighted translation game $H = \mathcal{G}$ with YES value $\geq 1 - d$, $d = k\zeta + p_*/2 \leq \varepsilon/t$, and NO value $\leq 99/100$. Corollary 2.3 (rounding, subdivision and t -fold repetition) gives the final instance: YES value $\geq 1 - t(d + \tau)/4 \geq 1 - \varepsilon/2$ and NO value $\leq \delta$, on the alphabet $\mathbb{F}_2^{\ell t}$, simple, bipartite, with translation constraints.

Running time. The game $\mathcal{G}$ has one group of constraints for each of the N^k equation tuples, where N is the number of equations, and a constant number (depending on $|\mathcal{V}|$ and m) of tables, noise values and vectors l per tuple. Keys are computed by linear algebra in fixed dimension. Rounding gives $O(h)$ edges and the t -fold product raises the size to a fixed power. All of this is polynomial for fixed ε, δ . $\square$

Remark 10.1 (A galactic polynomial). The exponent of the polynomial is at least k , and Lemma 8.1 needs $k\beta^2|\mathcal{V}| = k^{-1/3}|\mathcal{V}| \rightarrow 0$, so $k \gg |\mathcal{V}|^3$, where $|\mathcal{V}|$ is already a tower of exponentials in the gadget parameters. The reduction is polynomial time in the complexity-theoretic sense only. For the consequences below this is irrelevant: they are statements about NP-hardness.

11 Consequences

Theorem 1.1 turns the many results conditional on the UGC into NP-hardness results. Section 8 of the manuscript states the precise forms and checks the details needed to plug the theorem into each reduction. Among them:

- **Max-Cut:** approximating beyond the Goemans–Williamson ratio $\alpha_{\text{GW}} \approx 0.87856$ is NP-hard [9, 6].
- **Vertex Cover:** approximating within any factor below 2 is NP-hard [12].
- **Every Max-CSP:** beating the basic SDP integrality ratio is NP-hard [16].

- **Ordering CSPs:** beating a random ordering is NP-hard for every fixed ordering predicate, following Guruswami, Håstad, Manokaran, Raghavendra and Charikar.
- **Cuts and clustering:** Multicut, nonuniform Sparsest Cut, Min-2CNF[≡] Deletion and Correlation Clustering are NP-hard to approximate within every constant factor, following Chawla, Krauthgamer, Kumar, Rabani and Sivakumar.

The repository also contains direct proofs, not going through the UGC, of the Max-Cut, Vertex Cover, Min-UnCut and directed feedback vertex set results.

12 A guide for the careful reader

12.1 What this note does and does not establish

This note follows the manuscript’s proof and does not add new mathematics. In writing it I reworked the calculations reproduced above: the block lemmas and the alignment criterion, the probability that a block absorbs an update, the rank-loss and harmonic-potential bookkeeping, the enlargement step, the noise eigenvalue computation and spectral comparison, the folding argument for the answer set, the χ^2 estimate, the clean-coordinate reconstruction (including the intercept correction), and the final parameter arithmetic. I found no errors in these. I did not referee every step at the level of detail in the manuscript: the erasure argument’s concentration bound, the witness-transfer conditioning and the Fourier coset computation are presented here in condensed form, and the external theorems (Håstad, KMS, Dinur–Steurer) are taken as given. I did not run the Lean check (Section 1.5). If it passes, it is far stronger evidence than any reading.

12.2 Where to look hardest

1. **The gadget** (Section 3). This is the new idea and the one place where a conceptual error would sink everything. It is elementary and self-contained. Points worth checking: that the factorization of Lemma 3.9 uses a single γ for all children; that genericity must hold for *every* g because the descended lift may depend on the random isomorphism J ; and that the threshold $r_* = b + 2$ really is independent of ℓ .
2. **Using the inverse theorem uniformly.** Theorem 7.1 is applied to functions on very rectangular matrices (ℓ fixed, $m = 1 + 2k \rightarrow \infty$) and to functions that are not folded (after erasure). The quantifiers $\ell \geq \ell_0(\eta)$, $m \geq m_0(\eta, \ell)$ allow both.
3. **Abundance** (Lemma 7.3). The step from “one good slice exists” to “random advice is good with constant probability” is new relative to the 2-to-2 proof, and the reasoning summary shows the model identified it as a gap and repaired it.
4. **Exact keys** (Lemma 4.2). The intercept correction for third-position variables has to cancel exactly, and repeated equations across blocks must keep separate coordinates.
5. **Conditioning in Lemma 5.3.** The frozen data R must not include anything correlated with the clean questions beyond the clean mask, and each player must be able to rebuild its input from its own clean questions alone.

6. **Parameter order** (Table 1). In particular, t is chosen before the alphabet, which requires alphabet-independent repetition.

12.3 Correspondence with the manuscript

This note	Manuscript
Theorem 1.1	Theorem 1.1
Lemma 2.2, Corollary 2.3	Lemma 7.1, §7.2
Theorem 2.4, Lemma 2.5, (1)	Theorem 4.1, Lemma 4.2, Eq. (4.1)
Theorem 3.1	Lemma 3.1
Lemmas 3.2 to 3.4	Lemmas 3.2–3.4
Lemma 3.6	Definition 3.5, Lemma 3.6
Lemmas 3.7 and 3.9 and Proposition 3.10	Lemmas 3.7–3.8, Proposition 3.9
Section 3.7	§3.5
Lemma 4.2, (5)	§4.3
Lemma 4.4	Lemma 4.3
Definition 5.1, Proposition 5.2	Definition 5.2, Proposition 5.3
Theorem 7.1	Theorem 5.1
Lemma 6.1	Lemma 5.4
Lemmas 7.3 and 7.4	Lemma 5.5
Lemmas 8.1 to 8.3	Lemmas 5.6–5.8
Theorem 2.1, Lemma 5.3, Proposition 5.4	Theorem 6.1, Lemma 6.2, Proposition 6.3
Section 11	Section 8 and Appendix A

12.4 Notation

$K = \mathbb{F}_2^\ell$	the Unique Games alphabet before repetition
$\mathcal{V} \supseteq K, C, \mu$	latent space, nonlinear decoder, latent noise
p_*, r_*	gadget stability error and rank threshold
$U, O, \bar{\pi}$	equation tuple, sparse projection, answer projection
$L_U = \mathbb{F}_2^{1+2k}, e_0^\top$	homogeneous answer space, the t_0 coordinate
$P = (M, T)$	table, split into K -part (a matrix) and W -part
F_Q	restored answer of a labeling on question Q
η, α, r_s	inverse-theorem input, agreement and slice size
$A, S_O = AM_O$	advice map and advice rows
$\beta = k^{-2/3}$	projection rate

References

- [1] OpenAI. *The Unique Games Theorem*. Manuscript, September 23, 2026. <https://github.com/openai/math>.
- [2] M. Bellare, O. Goldreich, M. Sudan. Free bits, PCPs, and nonapproximability—towards tight results. *SIAM J. Comput.* 27(3), 1998.

- [3] B. Barak, P. Kothari, D. Steurer. Small-set expansion in shortcode graph and the 2-to-2 conjecture. ITCS 2019; arXiv:1804.08662.
- [4] I. Dinur, S. Khot, G. Kindler, D. Minzer, M. Safra. Towards a proof of the 2-to-1 games conjecture? STOC 2018.
- [5] I. Dinur, D. Steurer. Analytical approach to parallel repetition. STOC 2014.
- [6] M. Goemans, D. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *J. ACM* 42(6), 1995.
- [7] J. Håstad. Some optimal inapproximability results. *J. ACM* 48(4), 2001.
- [8] S. Khot. On the power of unique 2-prover 1-round games. STOC 2002.
- [9] S. Khot, G. Kindler, E. Mossel, R. O'Donnell. Optimal inapproximability results for MAX-CUT and other 2-variable CSPs? *SIAM J. Comput.* 37(1), 2007.
- [10] S. Khot, D. Minzer, M. Safra. On independent sets, 2-to-2 games, and Grassmann graphs. STOC 2017.
- [11] S. Khot, D. Minzer, M. Safra. Pseudorandom sets in Grassmann graph have near-perfect expansion. FOCS 2018; *Annals of Math.* 198(1), 2023.
- [12] S. Khot, O. Regev. Vertex cover might be hard to approximate to within $2 - \epsilon$. *J. Comput. Syst. Sci.* 74(3), 2008.
- [13] S. Khot, M. Safra. A two-prover one-round game with strong soundness. *Theory of Computing* 9, 2013.
- [14] D. Minzer, K. Zheng. Near optimal alphabet-soundness tradeoff PCPs. STOC 2024.
- [15] A. Rao. Parallel repetition in projection games and a concentration bound. *SIAM J. Comput.* 40(6), 2011.
- [16] P. Raghavendra. Optimal algorithms and inapproximability results for every CSP? STOC 2008.